



Applying the four operations to decimals

Teacher resource

Question 1

What is the value of $9.9 \div 0.3$?

0.033

☐

0.33

☐

3.3

☐

33

☐

Answer: D

Skill: Students use written methods for dividing decimals.

Background information

Students should be encouraged to develop and choose from a range of calculation strategies, including both formal and informal written methods. The particular strategy used should be chosen for efficiency and should depend on the specific numbers in the problem.

In this particular example, $9.9 \div 0.3$ could be thought of as $99 \div 3$, or 33.

Alternatively, 9.9 could be partitioned and thought of as $3 + 3 + 3 + 0.6$. This particular choice of partitioning lends itself to thinking of 9.9 as $(10 \times 0.3) + (10 \times 0.3) + (10 \times 0.3) + (3 \times 0.3)$ or 33×0.3 . These strategies can be used for mental calculation or, if needed, recorded in writing as informal written calculations.

Additional questions

1. What is the value of $8.8 \div 4$?
2. Evaluate $12.24 \div 0.12$.
3. Without evaluating the expressions $12.3 \div 4.1$ and $12.3 \div 0.41$, describe how the values of the expressions compare.

Question 2

A farmer has a 3.5 metre length of rope.
She cuts five pieces each 45 cm long from the rope.

Which of the following represents the length of rope left over?

- ☐ $(3.5 - 0.45) \times 5$
- ☐ $3.5 - (5 \times 0.45)$
- ☐ $3.5 \div 5 - 0.45$
- ☐ $3.5 - 5 - 0.45$

Answer: B

Skill: Students interpret expressions involving decimals in familiar contexts.

Additional questions

In terms of cutting pieces of rope from the 3.5 m length, describe what the following expressions could represent:

1. $3.5 - 0.45 - 2$
2. $(3.5 - 0.45)$

Question 3

A shop sells soft drink in single cans and in cartons of six cans.
Cartons of six cans cost less than six single cans.
A carton of six cans costs \$4.70.
Single cans cost 80 cents each.

Which of the following statements represents the amount saved by buying a carton of six cans rather than six single cans?

- ☐ $0.8 - (4.7 \div 6)$
- ☐ $4.7 - (6 \div 0.8)$
- ☐ $(6 \times 0.8) - 4.7$
- ☐ $6 - (4.7 \div 0.8)$

Answer: C

Skill: Students interpret the meaning of a numerical expression involving decimals in a familiar context.

Additional questions

1. In terms of the same situation, what does the expression $0.8 - (4.7 \div 6)$ represent?
2. In terms of the same situation, what does the expression $6 (4.7 + 0.8)$ represent?

Question 4

$$1.5 \times \Delta = 0.4 + 0.2$$

What is the value of Δ in the equation?

- ☐ $(0.4 + 0.2) \div 1.5$
- ☐ $1.5 \div (0.4 + 0.2)$
- ☐ $(0.4 + 0.2) \times 1.5$
- ☐ $1.5 \div (0.4 - 0.2)$

Answer: A

Skill: Students use the inverse relationship between multiplication and division to solve equations.

Additional questions

Write an equation, with an unknown variable, that can be solved by using the expression:

1. $(0.4 + 0.2) \times 1.5$
2. $1.5 \div (0.4 + 0.2)$

Student activity: Applying the four operations to decimals

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- ☐ $1.5 \div (0.4 - 0.2)$